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MILITARY SPECIFICATION

COMPRESSORS, RECIPROCATING OR ROTARY POWER DRIVEN (EMD), AIR, BASE MOUNTED, 10 HP TO 300 HP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers stationary, electric-motor-driven, reciprocating and rotary air compressors that operate in a power range of 10 horsepower (hp) to 300 hp.

1.2 Classification. Compressors shall be of applicable sizes, and of the following types, as specified (see 6.2.1):

Type I - Single acting (reciprocating).
Type II - Double acting (reciprocating).
Type III - Rotary (vane, screw).

* 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2.1), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

F-F-351 - Filters and Filter Elements, Fluid, Pressure: Lubricating Oil, Bypass and Full Flow.
GG-G-76 - Gages, Pressure and Vacuum, Dial Indicating (for Air, Steam, Oil, Water, Ammonia and Chloro-Fluoro Hydrocarbon Gases).
CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 Hp and Smaller).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704 - Treatment and Painting Material.
MIL-F-3541 - Fittings, Lubrication.
MIL-C-3600 - Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.

STANDARDS

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MS 35844 - Plug, Machine Thread, Magnetic: Drain.
MS 35875 - Air Cleaner, Intake-Oil-Bath Type, for Industrial Engines.
MS 39247 - Air Cleaner, Intake; Dry Type, for Internal Combustion Engines (class 1 and 11).
MS 49006 - Plugs, Pipe, Magnetic-headless; Aluminum, Iron or Steel.

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF LABOR (DoL)

Occupational Safety and Health Administration (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B15.1 - Mechanical Power Transmission Apparatus, Safety Standard for.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code - Section VIII Rules for Construction of Pressure Vessels.
ASME Power Test Code PTC 9 - Displacement Compressors, Vacuum Pumps and Blowers.

(Application for copies should be addressed to the American Society of Mechanical Engineers, P.O. Box 3199, Grand Central Station, New York, NY 10163.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ISI.1 - Controls and Systems, Enclosures for.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC., (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

RUBBER MANUFACTURERS ASSOCIATION (RMA)

Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturers Association, 44 Madison Avenue, New York, NY 10022.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The air compressor shall be either single or two-stage, and shall include a motor and controls as specified herein. The air compressor shall also include other components such as inter-cooler, oil cooler, after-cooler, receiver, and safety devices as required or as specified. The equipment shall be base mounted for stationary service, with sufficient mounting holes to provide firm anchorage to a foundation. When specified (see 6.2.1), the compressor and components shall be assembled into a packaged unit.

* 3.2 Standard commercial product. The equipment shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the equipment being furnished. A standard commercial product is a product which has been sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

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* 3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

* 3.4 First article. When specified (see 6.2.1), the contractor shall furnish a complete compressor for first article inspection and approval (see 4.3 and 6.3).

3.5 Identical items. Compressors of the same classification furnished under any specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable without prior written approval of the contracting officer.

3.6 Construction. The equipment shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

3.7 Ease of maintenance. The compressor and components shall be so designed and constructed that normal adjustments, repair, and overhaul can be readily accomplished by means of general purpose tools with a minimum removal or disturbance of other elements of the unit. Ease of maintenance provisions shall insure operation clearances for facilitating maintenance and servicing. Where possible, intricate locking devices, controls, and threaded fastenings that can be easily overtorqued by operators lacking feel through thick gloves or numbness, shall be avoided. Covers or plates that must be removed for component adjustment or for parts removable shall be equipped with substantial, quick-disconnect fastenings.

3.8 Performance. The compressor shall be capable of compressing and delivering not less than the specified volume flow rate of air, compressed and delivered at the specified pressure (see 6.2.1), under atmospheric conditions of 14.7 pounds per square inch (psi) pressure, 68° Fahrenheit (F) temperature, and 36 percent relative humidity. In addition, the power required by the compressor shall be calculated in accordance with ASME PTC 9 and shall be the rating indicated by the manufacturer with ±3 percent.

3.9 Compressor. The compressor shall be of the reciprocating or rotary-type, cable of operating unattended for a period of 4 hours at rated speed and capacity. The compressor shall be designed and constructed to withstand the normal strains, jars, vibrations, and other conditions incident to shipping, storage, and installation, so that no parts will work loose in service. All parts subject to wear or service shall be field replaceable, and the

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compressor shall require only minimum maintenance. Cover plates for access openings shall be oil- and dust-tight. The cylinder cooling area shall be adequate to prevent excessive heating. Air passages shall be unrestricted. The crankcase, oil reservoir, or receiver shall be provided with means of filling and draining oil and for checking the level of the oil. On two-stage reciprocating compressors, an intercooler shall be provided to cool the air between stages of compression (see 3.13). The compressor shall be equipped with an electric motor (see 3.17) and drive (see 3.19).

3.9.1 Type I, single acting. The type I compressor shall be a single-acting reciprocating type compressor with one or two stages as specified (see 6.2.1). The air temperature at the cylinder discharge flange shall not exceed 380°F, limited by a high-temperature shut-off switch. The compressor shall have air cooled cylinders unless water jacketed cooling is specified (see 6.2.1). Cylinder arrangement shall be vertical, radial, or V-type. When compressor is rated at 40 hp or greater, the cylinders shall be separate and individually detachable to make repairs simple and inexpensive. Pistons shall be of the automotive type, equipped with not less than four piston rings of which at least one shall be an oil ring. Lubrication shall be the splash type or forced feed system as described in 3.10, 3.10.1 and 3.10.2. Unless otherwise specified (see 3.15 and 6.2.1), the compressor, motor, and associated equipment shall be mounted on a substantial, single, unified frame assembly.

3.9.2 Type II, double-acting. The type II compressor shall be a reciprocating, double-acting, crosshead-type compressor with water-cooled cylinders and heads. The compressor shall be single- or two-stage as specified (see 6.2.1). The cylinder arrangement may be either horizontal, vertical, V-type, L-type, radial or semi-radial. Cylinder walls shall be of sufficient thickness to permit resleeving, reboring to a radial depth of at least 1/16 inch, or shall have removable, renewable cylinder liners. Pistons shall have at least two rings and be of minimum weight. Wearing surfaces on the crosshead shall be babbitted, adjustable, or replaceable unless similar surfaces are provided in the crosshead guides. Cylinder stuffing boxes shall be of sufficient depth to assure tightness against compression leakage without excessive piston rod pressure. Oil wiper rings at the frame end shall limit oil loss along the piston rod. The compressors shall be the lubricated type unless oil-free air is specified (see 6.2.1). When oil-free air delivery is specified, no part of the piston rod shall enter both the cylinder and the frame end, the piston rings shall be Teflon, and positive means, such as an oil deflection collar, shall be provided to ensure that no oil enters the cylinder via the piston rod. Lubrication shall be as specified in 3.10 and 3.10.2 except the running gear of horizontal compressors may be the splash type as specified in 3.10.1. Cylinder lubrication for all type II compressors, unless oil-free air is specified, shall be specified in 3.10.3.

3.9.3 Type III, rotary. The type III compressor shall be a single stage rotary-sliding vane or a symmetrical rotary-screw type compressor. The compressor shall be lubricated and cooled by the injection of filtered lubricating oil into the compression chamber. When the compressor is operated continuously at rated capacity (see 3.8), the temperature of the discharge air shall be not more than 100°F above the ambient air temperature. The compressor shall be equipped with an oil separator and receiver. The separator may be located within the receiver. The receiver shall be designed and constructed in accordance with the ASME Boiler and Pressure Vessel Code,

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Section VIII and shall be stamped with the official ASME code symbol. A spring-loaded safety valve marked with the ASME Code symbol shall be installed on the receiver. The receiver discharge system shall include an automatic blowdown valve which shall reduce the pressure in the receiver to a maximum of 10 psi by discharging air to atmosphere when the compressor is shutdown. A check valve shall be installed to prevent the backflow of system air into the receiver when the compressor is shutdown. The compressor shall automatically shutdown if the discharge air temperature exceeds the limits of the compressor's normal operating range. Compressors rated at 40 hp or greater shall operate at 30 percent of rated hp or less when operating unloaded. Lubrication for type III compressors shall be in accordance with 3.10 and 3.10.4.

3.9.4 Bearings. Unless otherwise specified (see 6.2.1), main bearings and connecting rod bearings may be of the type ordinarily employed by the manufacturer as his standard commercial practice.

3.9.5 Valves. Valves for all reciprocating compressors shall operate quietly and shall be readily removable for inspection.

3.9.6 Air intake. The air intake shall be suitably located to avoid the exhaust heat resulting from the compressor's cooling system. Airtight inlet piping shall be capable of resisting corrosion. The piping shall be free of sharp bends and shall be suitably routed so as not to interfere with the function and maintenance of adjacent components.

3.9.7 Air filter. The compressor air intake shall be fitted with an air filter in accordance with the manufacturer's latest standard practice. When specified (see 6.2.1), the air filter shall act as a muffler of the intake noises and minimize the restriction of air flow. The filter shall be of the oil-wetted type or dry type of equal efficiency and shall be readily removable for cleaning. When specified (see 6.2.1), the air filters shall be in accordance with either MS 39247 or MS 35875.

3.9.8 Safety valves. When no receiver is supplied, and unless otherwise specified (see 6.2.1), a safety valve shall be installed in the discharge piping of all compressors. A safety valve shall be installed in the discharge piping ahead of the first device in the discharge line between compressor stages. When the receiver is supplied, a safety valve shall be installed on the vessel. The safety valve(s) shall be the spring-loaded type, marked to show its capacity and pressure rating, and stamped with the ASME Code symbol. Both the safety valve(s) and its method of installation shall be in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII.

3.9.9 Discharge shut-off valve. When specified (see 6.2.1), the compressor discharge shall be equipped with the shutoff valve. The shutoff valve shall be installed downstream of the safety valve.

3.10 Lubrication. The lubrication system for each type of compressor shall include an oil level indicating device and an oil drain with plug. The oil filter, if required (see 3.10.2 and 3.10.4), shall be in accordance with the manufacturer's standard practice, except, when specified (see 6.2.1), the oil filter shall be either of the bypass or fullflow type in accordance with

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F-F-351. When specified (see 6.2.1), the drain plug shall conform to MS 49006 or MS 35844. When specified (see 6.2.1), the oil receiver, crankcase, or reservoir shall be equipped with an electric, thermostatically controlled immersion heater.

3.10.1 Splash type lubrication. The splash type lubrication system shall have an oil reservoir or crankcase from which the compressor running gear is lubricated. The compressor shall automatically shut off if the oil level gets too low.

3.10.2 Forced feed lubrication. The forced feed lubrication system for reciprocating compressor running gear shall include an oil reservoir or crankcase, a positive displacement pump, an oil filter, and a low oil pressure device for automatic shut off of the compressor. The positive displacement pump shall incorporate a pressure relief device to prevent excessive oil pressure from building up downstream of the pump. An oil cooler in accordance with the manufacturer's standard design shall be provided if the manufacturer normally supplies an oil cooler with the type of compressor specified.

3.10.3 Cylinder lubrication. Cylinder lubrication for type II compressors, unless oil-free as is specified, shall be provided by means of a forced-feed lubricator operated from the compressor running gear. The oil feed to each cylinder shall be visual and the oil feed regulation shall be adjustable.

3.10.4 Rotary compressor lubrication. The lubrication system for type III, rotary compressors, shall include an oil separator, receiver, cooler, and filter. The oil separator shall reduce the level of oil entrained in the discharge air to 15 parts per million or less. When a pressure in excess of 10 psi must be maintained in the receiver to ensure adequate circulation of compressor oil or to eliminate any carry-over, the air discharge piping for the receiver shall include a minimum pressure valve set to maintain a predetermined minimum back pressure on the compressor oil system. The oil cooler shall either be the water cooled shell-and-tube-type or the air-cooled fin-and-tube-type. The oil temperature of a shell-and-tube-type oil cooler shall be controlled by a thermostat that automatically regulates the flow of water or oil through the cooler. The lubrication system shall include a device for automatically stopping the compressor if the oil level or oil pressure becomes low. Check valves or other means shall be provided to prevent loss of oil through the intake filter.

3.11 Regulation. Regulation of the compressor shall be accomplished by one of the following two methods of control as specified (see 6.2.1). The regulator shall have adjustments for changing discharge and differential pressures.

3.11.1 Constant speed control. The compressor shall operate continuously at constant speed. Means shall be provided to automatically load and unload the compressor at preset minimum and maximum pressure settings respectively. Loading and unloading on two-stage double acting compressors shall be accomplished by a minimum of three steps (full load, one-half load, and no load). Means shall be provided for the automatic release of pressure within the cylinders when the unit is operating without load. Means shall also be provided for the manual or automatic unloading of the cylinders during the

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starting of the unit. When specified (see 6.2.1), the compressor shall be equipped with a timed control to stop the compressor after an adjustable unloaded period of up to 30 minutes if air is not used.

3.11.2 Dual control. The dual system shall consist of a combination of constant speed control and an automatic start-and-stop control by automatic or manual selector switch. When set for start-and-stop control, the motor shall stop automatically when the discharge pressure reaches the maximum pressure setting and start automaticaliy when the discharge pressure falls to the minimum setting. The cylinders shall unload immediately prior to shutdown and restarting and the cooling water shall shut off automatically during periods of motor shutdown.

3.12 Optional safety controls. In addition to controls specified elsewhere in this specification, the compressors shall be equipped with the optional safety controls listed in table 1 as specified (see 6.2.1). The controls shall provide an alarm of the appropriate condition by sounding a horn and by flashing an amber light or shall cause a shutdown of the compressor as indicated in the table for each type of compressor. Gages and visual identification lights for control points shall be panel-mounted or mounted on a separate working-height console as specified (see 6.2.1).

TABLE 1. Optional safety controls.

	Type I		Type II		Type III	
Control point	Alarm	Shut Down	Alarm	Shut Down	Alarm	Shut Down
High-lubrication oil temp.	-	-	x	x	x	x
Cylinder lubrication failure	-	-	-	x	-	-
High main bearing temperature	-	x	-	x	-	x
High discharge-air temperature	x	x	x	x	x	x
High discharge-air pressure	x	x	x	x	x	x
High intercooler air temp.	x	x	x	x	-	-
Low cooling water flow	-	x	-	x	-	-
Low water pressure	-	x	-	x	-	-
High water temperature	x	x	x	x	-	-
High intercooler moisture separator level (two-stage compressors only)	x	x	x	x	-	-
Excessively high motor temp.	x	-	x	-	x	-
Excessive vibration	-	x	-	x	-	-

3.13 Intercoolers and aftercoolers. Intercoolers shall be supplied with all two stage compressors and aftercoolers shall be supplied when specified (see 6.2.1). Intercoolers for air-cooled compressors shall be the tube-and-fin-type. Intercoolers for water-cooled compressors shall be the shell-and-tube-type, except that tube-and-fin-type may be used when the intercooler is supported by the compressor frame or attached to the compressor. Air or water cooled intercoolers may be the integral cast type when compressor hp is 25 hp or less. Aftercoolers shall be of the water-cooled shell-and-tube-type or air-cooled tube-and-fin-type. Water cooled aftercoolers and intercoolers shall be of sufficient capacity to cool the compressed air to within 15°F and 20°F, respectively, of the temperature of the water entering the cooler. Air cooled intercoolers and aftercoolers shall have sufficient capacity to cool the compressed air to within 20°F of the ambient temperature under the

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atmospheric conditions specified in 3.8. Water cooled intercoolers and aftercoolers shall be provided with sight-flow indicator to visually observe the flow of water to the cooler. The pressure drop of compressed air through the cooler shall not exceed 1 psi for operating pressures above 70 psi or 2.5 psi for operating pressures equal to or below 70 psi. All intercoolers and aftercoolers shall be provided with a moisture separator and drain trap to remove the condensed moisture and oil from the air leaving the cooler.

3.13.1 Shell-and-tube. Shell-and-tube coolers shall be of the floating-head type consisting of a removable and cleanable nest of corrosion-resistant tubes within a steel shell. Air may pass either through the tubes or the shell.

3.13.2 Tube-and-fin. Tube-and-fin coolers shall be copper, aluminum, copper-aluminum, or copper-alloy construction. Fins shall be securely bonded to the tubing. Tube-and-fin coolers shall be provided with a fan for circulation of the cooling air. The fan shall be adequately guarded for safety and be driven either from the compressor crankshaft or by an independent electric motor.

3.14 Gages. Dial type pressure indicating gages shall be provided with all compressors to indicate discharge or receiver air pressure, air pressure between stages for two-stage compressors 25 hp and larger, and oil pressure for pressurized or force feed lubrication system. Gages shall be in accordance with GG-G-76 and shall be mounted on the equipment.

3.15 Air receiver. An air receiver shall be furnished when specified (see 6.2.1). Unless specified otherwise (see 6.2.1), the volume of the receiver shall not be less than shown on table II. The receiver shall be designed and constructed in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII and shall be stamped with the official code symbol. The receiver shall be designed for a working pressure 10 psi greater than the maximum working pressure of the cylinder. The receiver shall be equipped with a safety valve as specified in 3.9.8 and a drain valve. When specified (see 6.2.1), a stand for mounting the receiver in the vertical or horizontal position shall be provided. When specified (see 6.2.1), and compressor hp is 30 hp or less and an air receiver is specified, the compressor and motor shall be mounted on the receiver.

TABLE II. Minimum receiver volume.

Compressor capacity Cubic feet per minute Standard Air	Minimum receiver volume Cubic Feet
45	4.5
80	8 (60 gal.)
110	11 (80 gal.)
125	13.5 (100 gal.)
150	16 (120 gal.)
190	19 (150 gal.)
250	27 (200 gal.)
300	32 (240 gal.)
340	34
500	53.5 (400 gal.)
570	57
960	96
2115	151

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3.16 Housing. When specified (see 6.2.1), the type III compressor shall be provided with a sheet metal housing which shall enclose the motor, compressor, and instrument panel. The housing shall include quick opening side panels or hinged access openings to the motor, compressor, and instrument panel. When required, the housing shall include grills that permit maximum passage of cooling air for the air-cooled radiator and oil cooler of the compressor. Sheet metal for the housing and panels shall be steel not less than 0.0598-inch (US revised standard gage No. 16) thick. Supporting members and braces shall be furnished as required to insure rigidity and to minimize vibrational noise during operation. Panels shall be sized to provide maximum accessibility to the compressor components. Upward-opening hinged access panels shall be equipped with devices to hold the panels in an open position. Removable access covers used to facilitate accessibility to isolated components shall be equipped with maintenance free quick-disconnect fasteners. Housing shall be sound attenuated to meet OSHA standards.

3.17 Motors. Unless otherwise specified (see 6.2.1), the compressor motor shall conform to CC-M-1807 and shall be suitable for continuous duty. The motor shall be designed for operation with electrical power supply having characteristics as specified (see 6.2.1). The motor shall have a hp nameplate rating equal to or greater than 90 percent of the hp required for continuous operation of the compressor at full load. Dripproof motors shall be poly-phase induction type. When specified (see 6.2.1), motors 125 hp and larger shall be the synchronous type. Motors that drive auxiliary components shall conform to CC-M-1807 and shall have electrical power supply characteristics as required.

3.18 Motor starter. Unless otherwise specified (see 6.2.1), a starter shall be furnished with capacity and electrical characteristics suitable for the motor according to NFPA No. 70. When specified (see 6.2.1), motors rated above 50 hp shall have reduced voltage, partial winding, or other means to limit starting current inrush to 200 percent of the normal full-load value. The starter shall include thermal overload protection and a NEMA type 1 enclosure as defined by 1S1.1.

3.18.1 Starting controls - Air Force requirements. Control circuits for motors above 20 hp shall be nominal 110 volt. Spare thermal protector units shall be securely fastened inside the starter enclosure.

3.19 Drive. The compressor shall be driven by a multi-V-belt drive or a direct drive as specified (see 6.2.1). When a reciprocating compressor is connected to a synchronous motor, a direct drive shall be provided. Direct drive units shall either be of the close-coupled, integral-coupled, or flexible-coupled type as specified (see 6.2.1). Flexible couplings shall be guarded. Couplings for direct drive compressors shall be designed to operate safely at 150 percent of the motor hp rating. Drivers and belts for belt driven compressors shall conform to RMA Engineering Standard for Multiple V-Belt Drives. Belts shall be designed with a service factor of at least 1.5. Belt-driven units shall have means for adjusting belt tension. Belt guards shall conform to ANSI B15.1.

* 3.20 Electromagnetic interference characteristics. When specified (see 6.2.1), the compressor shall conform to the electromagnetic interference suppression requirements and test limits of SAE J551.

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3.21 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

* 3.22 Cleaning, treatment, and painting. Unless otherwise specified (see 6.2.1), surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. When specified (see 6.2.1), the unit shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The color of the finish coat shall be as specified (see 6.2.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants, such as soil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.23 Lubricants. Lubricants shall be in accordance with the equipment manufacturer's recommendations and shall be compatible with lubricants specified herein. Pressure lubrication shall not damage seals or other parts. Unless specified otherwise (see 6.2.1), all parts requiring lubrication shall be lubricated prior to delivery and tagged to show the type and temperature rating of the lubricant used. The equipment shall be designed to operate efficiently and satisfactorily when lubricated with standard military lubricants (see 6.4).

3.23.2 Lubrication fittings. Lubrication fittings shall conform to types 1, 11, and 111 of MIL-F-3541 and shall be located in accessible, protected positions. A bright red circle shall be painted around each lubricating point.

3.24 Skid base. When specified (see 6.2.1), type 11 and type 111 compressors shall be mounted on a steel skid base. The base shall consist of two steel skid runners with an integrally formed, welded, or bolted base plate for mounting components of the unit. The skid runners shall be upturned or beveled at each end to permit skidding the unit into place. The skid runners shall be furnished with mounting holes for anchoring the units to a concrete foundation or wooden deck. The number and size of holes required shall be as determined by the manufacturer for anchorage of the unit under the most severe condition of vibration or thrust likely to develop during operation. The skid runners shall be of sufficient height to provide ready access to any connections located under the units and shall be of sufficient width to provide adequate bearing surface on a smooth gravel base. Individual mountings shall be furnished on the base plate as required for supporting motors, pumps, and other components in vertical alinement. Other components requiring removal for service, adjustment, or horizontal alinement shall be mounted in a manner to permit such removal or alinement. Lifting attachments and tiedown devices, if furnished (see 3.25), shall be welded to the skid base.

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3.25 Lifting and tying down attachments. When specified (see 6.2.1), the compressor shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to MIL-STD-209, type III. Attachments may be identified by stenciling or other suitable marking on the compressor. Tying down marking shall clearly indicate that the attachments are intended for tying down of the compressor on the carrier when shipped.

3.26 Identification plates. The compressor shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.28 Spare parts and maintenance tools. When specified (see 6.2.1), spare parts and maintenance tools shall be furnished. Special tools shall be kept to a minimum.

3.28.1 Toolbox. When specified (see 6.2.1), a toolbox shall be provided. The toolbox shall be large enough to store specialized tools not carried in mechanics toolbox, and required for field service or maintenance, but shall not have external closed dimensions less than 14 inches in length, 6 inches in width and 6 inches in height. The toolbox shall be of nominal 0.0747-inch (U.S. revised standard gage No. 14) thick steel with a hinged lid and a trunk drawbolt to keep the lid secure when vibrated. The toolbox shall be mounted in a protected, accessible location.

* 3.29 Workmanship.

3.29.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.29.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.29.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.29.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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3.29.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on one compressor when first article sample is required (see 3.4 and 6.3). This inspection shall include the examination of 4.5, the tests of 4.6 and when specified (see 4.7 and 6.2) a preproduction pack inspection. The first article may be a standard production item from the contractor's current inventory provided the compressor meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining compressors to be furnished under the contract.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on the compressors. This inspection shall include the examination of 4.5, the tests of 4.6.1, 4.6.2, and 4.6.4, and the packaging inspection of 4.7.

4.5 Examination. Each compressor shall be examined for compliance with the requirement specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Tests. Each compressor shall be tested, and any unit failing to pass the following tests, as applicable, shall be rejected. Certified test reports from an independent laboratory approved by the contracting officer will be accepted as evidence that the unit has passed the required tests. Tests shall be conducted as outlined in the referenced documents as herein specified. Prior to the tests specified herein which require the compressor(s) to be operated, the unit(s) shall be serviced with the oils and greases recommended by the manufacturer for use in the ambient temperature at the test location.

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4.6.1 Receiver test. The receiver shall be hydrostatically tested in accordance with the requirements of the ASME Boiler Pressure Vessel Code, Section VIII and held at this pressure until examined for leaks and distortion. Leaks or distortion shall be cause for rejection. The ASME Code symbol stamp will be accepted as evidence of this test.

4.6.2 Safety valve. The safety valve shall be tested to determine if its relieving capacity and popping pressure are in accordance with the requirements of the ASME Boiler and Pressure Vessel Code, Section VIII.

4.6.3 Performance test. The compressor shall be tested to determine capacity and power consumption in accordance with the rules, procedures, and instrumentation specified in the ASME PTC 9. The duration of the test shall be sufficient to record all data required under Report of Test of PTC 9, but in no case shall the test duration be less than 1 hour. Test results shall be corrected to the atmospheric conditions specified in 3.8. Prior to the actual performance test, preliminary tests shall be run, as required, to verify that the compressor and associated systems are in suitable condition, that regulation controls have been properly adjusted, and that gages and instruments are functioning properly. Inability of the compressor to meet the performance requirements of this specification or to operate satisfactory shall be cause for rejection.

4.6.4 Operational test. Each compressor, excluding the unit offered for first article inspection, shall be tested for not less than 15 minutes to verify satisfactory operation of the compressor and all controls. During the test, all necessary adjustments in controls shall be made as required to insure that performance requirements relating to the compressor output and regulation shall be met.

4.6.5 Test for electromagnetic interference characteristics. When electromagnetic interference characteristics are required, the compressor shall be tested to determine conformance to 3.20. In lieu of tests to determine conformance to SAE J551, the manufacturer may furnish a certification that the compressor meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.7 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of Section 4 of MIL-C-3600. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2.1).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-C-3600 with the level of preservation and the level of packing as specified (see 6.2.1).

6. NOTES

6.1 Intended use. This specification covers air compressors intended for stationary service in installations where a steady volume of air in a pressure range of 5 to 125 psi is required.

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6.1.1 Type I. Type I should be specified for applications that require light weight, compactness, low initial investment, and power requirements not exceeding 125 hp.

6.1.2 Type II. Type II should be specified for continuous heavy-duty service when low operating costs and minimum attention to operation and maintenance are important factors.

6.1.3 Type III. Type III should be specified for applications that require light weight, compactness, continuous duty service, and minimum attention to operation and maintenance.

6.2 Ordering data. Acquisition documents should specify:

6.2.1 Acquisition requirements:

a. Title, number, and date of this specification.
b. Type of unit required (see 1.2).
c. When specifications and standards shall be other than as specified (see 2.1.1).
d. When the compressor and components shall be assembled into a packaged unit (see 3.1).
e. When first article is required (see 3.4, 4.5 and 6.3).
f. The capacity (volume flowrate of air in cubic feet per minute) and the discharge working pressure required (see 3.8).
g. Number of stages required for type I compressor and if water jacketed cooling is required (see 3.9.1).
h. Number of stages required for type II compressor (see 3.9.2).
i. When oil-free air delivery is required for type II compressors (see 3.9.2).
j. Type of bearings, if other than manufacturer's standard bearing (see 3.9.4).
k. When air filters shall act as a muffler (see 3.9.7).
l. When air filters shall be in accordance with MS 39247 or MS 35875 (see 3.9.7).
m. When a safety valve is not to be installed in the compressor discharge piping when no receiver is supplied (see 3.9.8).
n. When a shutoff valve is required on the compressor discharge (see 3.9.9).
o. When oil filter shall be in accordance with F-F-351 and whether it shall be by-pass or full-flow (see 3.10).
p. When drain plug shall conform to MS 49006 or MS 35844 (see 3.10).
q. When an electric, thermostatically controlled immersion heater is required in the oil receiver or crankcase (see 3.10).
r. Required method of compressor regulation (see 3.11).
s. When timed controls are required to stop compressor (see 3.11.1).
t. Which additional optional safety controls are required (see 3.12).
u. Whether gages and visual identification lights shall be panel-mounted or mounted on a working-height console separate from the compressor (see 3.12).
v. When aftercoolers are required (see 3.13).
w. When an air receiver is required (see 3.15).
x. When receiver volume shall be other than specified (see 3.15).
y. When stand is required for receiver and whether intended for horizontal or vertical mounting (see 3.15).

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z. When the compressor and motor are to be tank mounted (see 3.9.1 and 3.15).
aa. When housing is required (see 3.16).
bb. When motor is other than that specified (see 3.17).
cc. The required electrical power supply characteristics for motors (see 3.17).
dd. When motors of 125 hp and larger shall be synchronous type (see 3.17).
ee. When motor starter is to be other than specified (see 3.18).
ff. When reduced voltage, partial winding or other means of limiting current inrush is required (see 3.18).
gg. If compressor shall be multi-V-belt or direct-driven and, if direct-driven, whether drive is to be close-coupled, integral-coupled, or flexible-coupled (see 3.19).
hh. When electromagnetic interference characteristics required (see 3.20).
ii. When fungus resistance is required (see 3.21).
jj. When cleaning, treatment, and painting is other than specified (see 3.22).
kk. When MIL-T-704, type A, treatment and painting is required (see 3.22).
ll. Type of color of the finish coat is required (see 3.22).
mm. When initial lubrication service shall be other than specified (see 3.23).
nn. When skid bases are required (see 3.24).
oo. When lifting and tiedown attachments are required (see 3.25).
pp. When spare parts and maintenance tools shall be furnished (see 3.28).
qq. When toolboxes are required (see 3.28.1).
rr. When a production pack inspection is required and the time frame required for submission (see 4.7).
ss. Level of preservation and level of packing required (see 5.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete compressor, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for examinations, tests, and approval of the first article.

6.4 Standard lubricants. When requested by the contractor, the contracting officer should furnish a list of standard Military lubricants applicable to the equipment covered by this specification. Standard Military lubricants are listed in the DoD section of the Federal Supply Catalog C9100-IL.

* 6.5 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

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* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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